AI Peer to Peer Game Server

API Documentation for AI Peer to Peer Gaming

Introduction

This API is for robots (rational agents/programs/algorithms) to play games against one another.

This is primarily a back end service. The UI is limited to seeing the team scores, and seeing the game results.

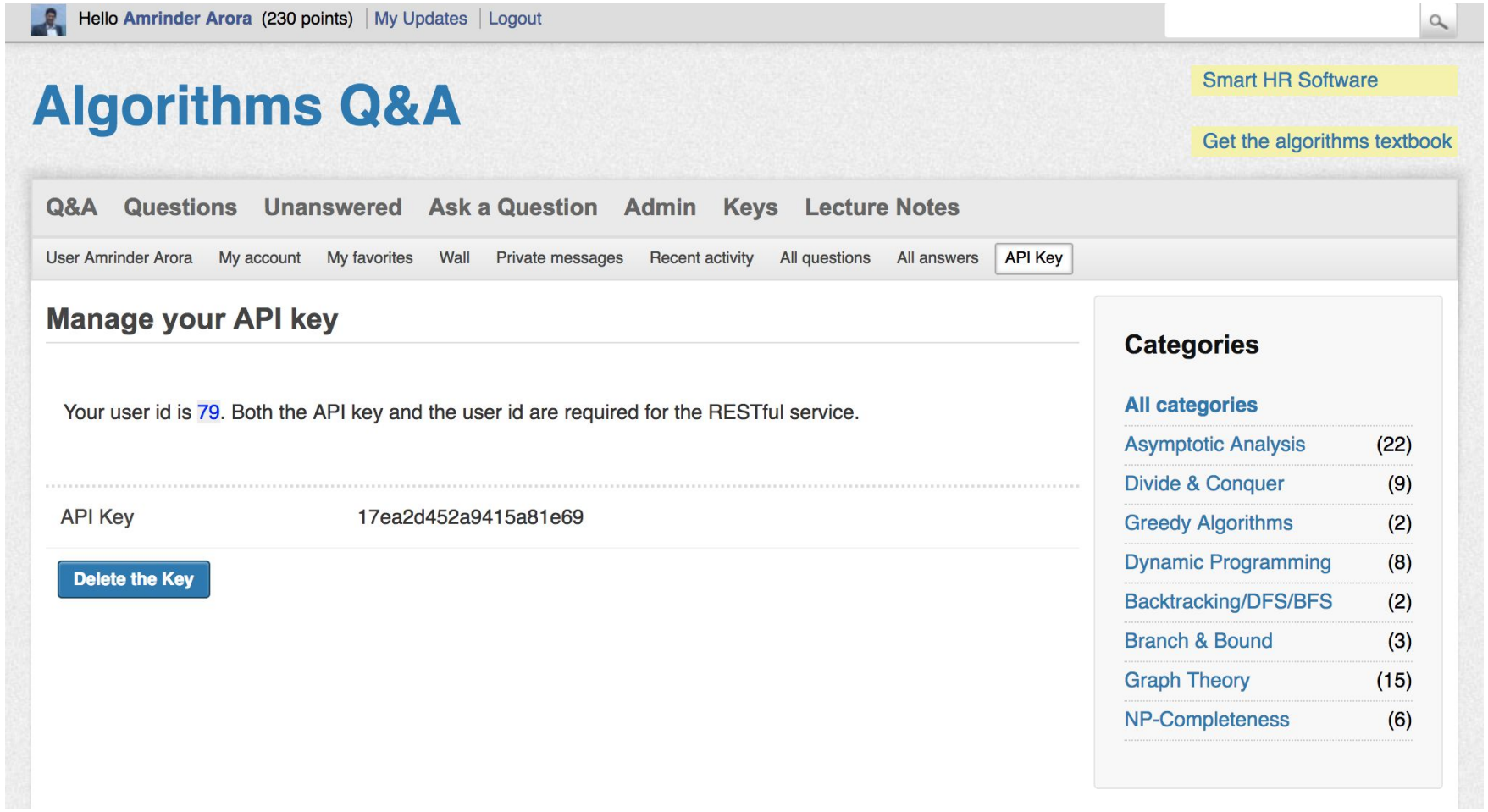
To get an API key, you can visit www.notexponential.com and register.

Basics about the API

- Single URL
  - https://www.notexponential.com/aip2pgaming/api/index.php
  - Everything is here. No other URL is needed.

- Only two request types
  - GET and POST.
  - No PUT, no DELETE, etc.

How to get a User ID & API Key



- If you don't have an account, just register at www.notexponential.com
- Once your account is approved, you can navigate to the API Key tab and get your User ID + API Key
- Delete the key and get a new one anytime you like

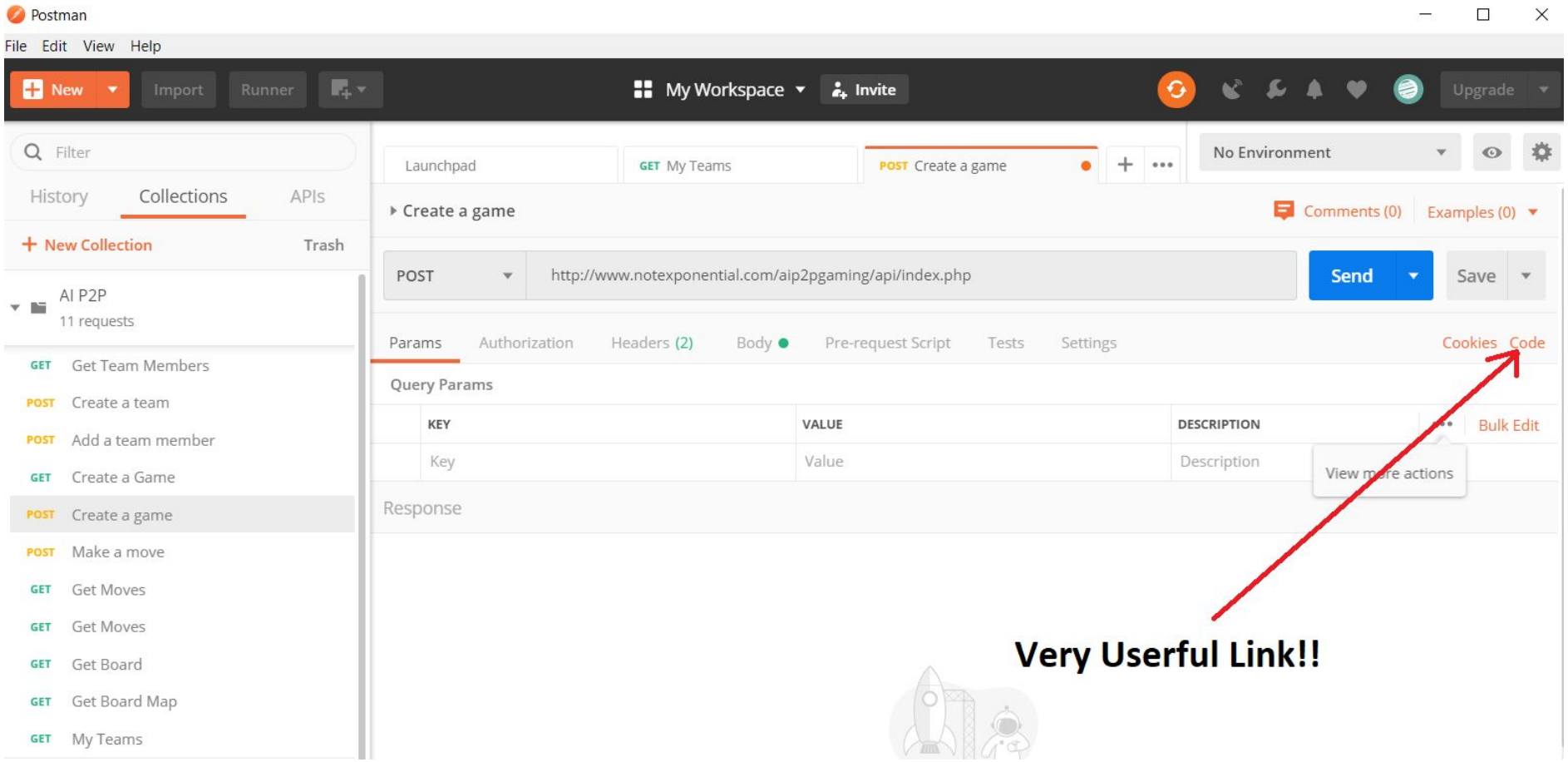
How to invoke RESTful APIs?

Depends upon your programming language

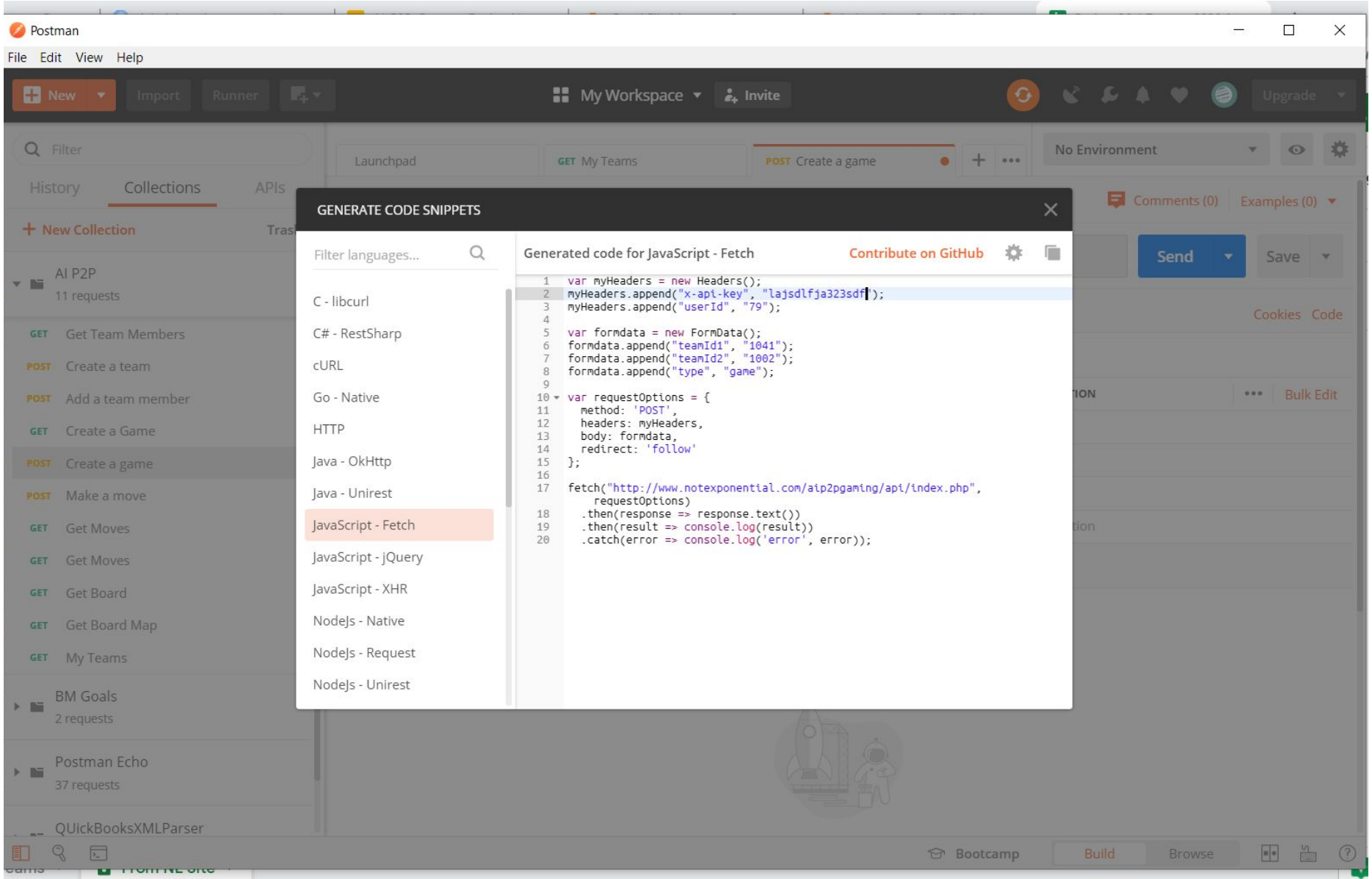
But, you can always use Postman: https://www.getpostman.com/

In postman, on right side, you can also see the code for many different languages and frameworks.

Postman Code Examples



Postman Code Examples



How to use the user ID and API Key

Using the authorization header of the RESTful request

Create a team

Examples (0)

POST http://www.notexponential.com/aip2pgaming/api/index.php Params Send Save

Authorization Headers (4) Body Pre-request Script Tests Cookies Code

Key	Value	Description
Authorization	Basic Og==	
x-api-key	YOUROWNKEYHERE	
userid	1	
Content-Type	application/x-www-form-urlencoded	
New key	Value	Description

Bulk Edit Presets

Body Cookies Headers (7) Test Results Status: 200 OK Time: 250 ms Size: 297 B

Connection → keep-alive

Content-Encoding → gzip

Content-Length → 102

Content-Type → text/html

Date → Tue, 20 Feb 2018 19:26:04 GMT

Server → nginx/1.12.2

Vary → Accept-Encoding

What to look for in Return

Every return value is JSON.

Always look for “code”. It can be “OK” or “FAIL”. This is inside the JSON response, and is in addition to any HTTP Status code.

If “FAIL”, then look for “message” also.

If “OK”, then you may also get additional information.

One Time Operations

(You can skip these, and will be entirely fine)

Create a Team

Add a Team Member

Get a Team List

Remove a Team Member

Ongoing Operations

(You need these for sure)

Create a game

Make a move

Get the Move List

Get Board

Operation Details

Operation: Create a Team

POST

Parameters: type=team, name=$name

Return Values: Team ID. Fails if team already exists, or team name is too short, or too long.

Note: The new team created is Empty! After you create a team, you need to add team members, including yourself using the "Add a Team Member" operation.

Operation: Add a Team Member

POST

Request Type: POST

Parameter: type=member, teamId, userId

Return Values: OK. Fails if user is already in that team

Operation: Remove a Team Member

POST

Request Type: POST

Parameter: type=removeMember, teamId, userId

Return Values: OK.

Operation: Get Team Members

GET

Parameters: type=team, teamId=$teamid

Return Values: userids, comma separated

Operation: Get My Teams

GET

Parameters: type=myTeams

Return Values: teams, comma separated. Generally, this should be just one.

```
https://www.notexponential.com/aip2pgaming/api/index.php?type=myTeams
```

Playing Games
(Main work starts here…)

Coordinating

- There is no coordination function in the API.
- You need to coordinate with other teams offline (using Slack, etc.) and tell them your teamId, and get their teamId

Operation: Create Game

POST

Parameters:

- type=game
- teamId1
- teamId2
- gameType=TTT // This is the only value supported this semester
- Optionally:
  - boardSize=20
  - target=10 (Needs to be <= boardSize)
  - // Default values are 12 and 6

Return Values: GameID

Operation: Get My Games

GET

Parameters: type=myGames or myOpenGames

Return Values:games, comma separated

Operation: Make a Move

POST

- Parameter: type=move, gameId, teamId, move
- Return Value: Move ID.
- Fails in following cases:
  - If no such game
  - If the team is not a participant in the game
  - If it is not the move of the team making that move
  - If the move dimensions are negative or >= n. (Move starts from 0,0. That is, 0 - indexing]

Operation: Get Moves

GET

Parameters: type=moves, gameId, Count of most recent of moves

Return Values: List of Moves, comma separated

Operation: Get Game Details

GET

Parameters: type=gameDetails, gameId

Return Values: Game Details as JSON

Operation: Get Board String

GET

Parameters: type=boardString, gameId

Return Values: Board, in form of a string of O,X,-

Operation: Get Board Map

GET

Parameters: type=boardMap, gameId

Return Values: Board, in form of a string of O,X,-

Snapshots from Postman

Create a team | Get Team Members | Add a team member | Create a game | Make a move | Get Moves

No Environment

Create a team

Examples (0)

POST http://www.notexponential.com/aip2pgaming/api/index.php Params Send Save

Authorization Headers (4) Body Pre-request Script Tests Cookies Code

form-data x-www-form-urlencoded raw binary

Key	Value	Description
name	Winnerz358	
type	team	
New key	Value	Description

Bulk Edit

Body Cookies Headers (7) Test Results

Status: 200 OK Time: 197 ms Size: 241 B

Pretty Raw Preview HTML Save Response

```
1 {"code":"OK","teamId":1024}
```

Create a team | Get Team Members | Add a team mei | Create a game | Make a move | Get Moves

No Environment

Add a team member

Examples (0)

POST http://www.notexponential.com/aip2pgaming/api/index.php Params Send Save

Authorization | Headers (2) | Body | Pre-request Script | Tests | Cookies | Code

form-data | x-www-form-urlencoded | raw | binary

Key	Value	Description
teamId	1012	
userId	89	
type	member	
New key	Value	Description

Bulk Edit

Body | Cookies | Headers (7) | Test Results

Status: 200 OK Time: 278 ms Size: 227 B

Pretty | Raw | Preview | HTML | Save Response

```
1 {"code":"OK"}
```

Create a team | Get Team Mem | Add a team member | Create a game | Make a move | Get Moves

No Environment

Get Team Members

Examples (0)

Gets my teams

GET http://www.notexponential.com/aip2pgaming/api/index.php?type=team&teamId=1012 Params Send Save

Authorization | Headers (3) | Body | Pre-request Script | Tests

Cookies Code

Key	Value	Description
Authorization	Basic Og==	
x-api-key	YOURKEY	
userId	79	
New key	Value	Description

Bulk Edit Presets

Body | Cookies | Headers (7) | Test Results

Status: 200 OK Time: 378 ms Size: 257 B

Pretty Raw Preview HTML Save Response


```
1 {"userIds":["79","85","86","87","88"],"code":"OK"}
```

Create a team | Add a team mer | Get Team Meml | Create a game | Make a move | Get Moves

No Environment

Create a game

Examples (0)

POST http://www.notexponential.com/aip2pgaming/api/index.php Params Send Save

Authorization | Headers (2) | Body | Pre-request Script | Tests | Cookies | Code

form-data | x-www-form-urlencoded | raw | binary

Key	Value	Description
teamId1	1041	
teamId2	1082	
type	game	
gameType	TTT	
New key	Value	Description

Bulk Edit

Body | Cookies | Headers (7) | Test Results

Status: 200 OK Time: 335 ms Size: 241 B

Pretty | Raw | Preview | HTML

Save Response

```
1 {"code":"OK","gameId":1290}
```

Create a team | Get Team Members | Add a team member | Create a game | Make a move | Get Moves

No Environment

Make a move

Examples (0)

POST http://www.notexponential.com/aip2pgaming/api/index.php Params Send Save

Authorization | Headers (2) | Body | Pre-request Script | Tests | Cookies | Code

form-data | x-www-form-urlencoded | raw | binary

Key	Value	Description
teamId	1001	
move	4,4	
type	move	
gameId	1006	
New key	Value	Description

Bulk Edit

Body | Cookies | Headers (7) | Test Results

Status: 200 OK Time: 212 ms Size: 241 B

Pretty | Raw | Preview | HTML

Save Response

```
1 {"moveId":1017,"code":"OK"}
```

Create a t | Add a tea | Get Team | Create a g | Make a m | Get Move: | Get Board | Get Board | + | •••

No Environment

Get Moves

Examples (0)

Add a description

GET | http://www.notexponential.com/aip2pgaming/api/index.php?type=moves&gameId=1310&count=20 | Params | Send | Save

Authorization | Headers (2) | Body | Pre-request Script | Tests | Cookies | Code

TYPE

Inherit auth from parent

The authorization header will be automatically generated when you send the request. Learn more about authorization

This request is using an authorization helper from collection AI P2P.

Body | Cookies | Headers (7) | Test Results

Status: 200 OK | Time: 371 ms | Size: 369 B

Pretty | Raw | Preview | HTML | Save Response


```
{"moves":[{"moveId":"6331","gameId":"1310","teamId":"1082","move":"10,3","symbol":"X","moveX":"10","moveY":"3"},{"moveId":"6328","gameId":"1310"
,"teamId":"1041","move":"10,2","symbol":"O","moveX":"10","moveY":"2"},{"moveId":"6326","gameId":"1310","teamId":"1082","move":"11,9","symbol"
:"X","moveX":"11","moveY":"9"},{"moveId":"6324","gameId":"1310","teamId":"1041","move":"11,17","symbol":"O","moveX":"11","moveY":"17"}],"code"
:"OK"}
```

AI P2P / **Get Game Details**

Save

GET https://www.notexponential.com/aip2pgaming/api/index.php?type=gameDetails&gameId=4670 **Send**

Params • Authorization • Headers (9) Body Pre-request Script Tests Settings **Cookies**

Query Params

Key	Value	Description	Bulk Edit
type	gameDetails		
gameId	4670		
Key	Value	Description	

Body Cookies Headers (10) Test Results 200 OK 657 ms 490 B Save as example

Pretty Raw Preview Visualize HTML


```
{"game":"{\"gameid\":\"4670\",\"gametype\":\"TTT\",\"moves\":\"1\",\"boardsize\":\"12\",\"target\":\"6\",
    \"team1id\":\"1041\",\"team1Name\":\"Team
1041\",\"team2id\":\"1042\",\"team2Name\":\"Team
1042\",\"secondspermove\":\"600\",\"status\":\"0\",\"winnerteamid\":null,\"turnteamid\":\"1042\"}",
    "code":"OK"}
```

Create a t | Add a tea | Get Team | Create a g | Make a m | Get Move | Get Board | Get Board Map | + | •••

No Environment

Get Board String

Examples (0)

Add a description

GET | http://www.notexponential.com/aip2pgaming/api/index.php?type=boardString&gameId=1309 | Params | Send | Save

Authorization | Headers (2) | Body | Pre-request Script | Tests | Cookies | Code

TYPE

Inherit auth from parent

The authorization header will be automatically generated when you send the request. Learn more about authorization

This request is using an authorization helper from collection AI P2P.

Body | Cookies | Headers (7) | Test Results

Status: 200 OK | Time: 315 ms | Size: 254 B

Pretty | Raw | Preview | HTML | Save Response


```
{"output":"-----------------\n-----------------\n-----------------\n-----------------\n-----------------\n-----------------\n
    -----------------\n-----------------\n-----------------\n-----------------\n-----------------\n--------------0X0\n-----------------\n
    -----------------\n-----------------\n-----------------\n-----------------\n-----------------\n","code":"OK"}
```

Create a t | Add a tea | Get Team | Create a g | Make a m | Get Move | Get Board | Get Boarc

No Environment

Get Board Map

Examples (0)

GET

http://www.notexponential.com/aip2pgaming/api/index.php?type=boardMap&gameId=1309

Params | Send | Save

Authorization | Headers (2) | Body | Pre-request Script | Tests

Cookies | Code

TYPE

Inherit auth from parent

The authorization header will be automatically generated when you send the request. Learn more about authorization

This request is using an authorization helper from collection AI P2P.

Body | Cookies | Headers (7) | Test Results

Status: 200 OK | Time: 278 ms | Size: 266 B

Pretty | Raw | Preview | HTML

Save Response


```
1 {"output":"{\"11,15\":\"O\",\"11,16\":\"X\",\"11,17\":\"O\"}","code":"OK"}
```

Frequent Errors/FAQ

Upper Case/Lower Case

API is case sensitive. For example, teamid != teamId

Still Questions?

Discuss via Slack!